

Inter-Annotator Agreement: Cohen’s Kappa, Worked Out by Hand

Why “83% of our annotators agreed” is a lie until you subtract luck — one 2×2 agreement table, every term computed.

What this document does. It takes the one tiny annotation study from Module 3.5 — two reviewers each labelling a batch of agent trajectories *acceptable?* (good/bad) — fixes a concrete 2×2 agreement table over $N = 50$ items, and then computes the observed agreement P_o , the chance agreement P_e , and Cohen’s κ *by hand*, term by term. Nothing is asserted. Each number below is reproduced by the small Python program in Appendix A, so the arithmetic is exact and self-consistent. We land at $\kappa = 0.70$ — “substantial” on the Landis–Koch scale the module quotes — then redo it for three categories, and finish on the *kappa paradox*: a table with 92% raw agreement whose κ is only 0.29.

Where this sits. This is **Topic 1 of Module 3.5** (Human Evaluation Protocols) — the gate you check *before* you publish any human-eval number. Its companions: Topic 2 of the same module (win-rate confidence and sample size — the *other* honesty check on a human study) and Module 3.2 (LLM-as-judge, where κ also scores judge-vs-human agreement).

Concepts covered, in order: the agreement table as a confusion matrix · observed agreement $P_o = \text{diag}/N$ · marginal label rates · chance agreement $P_e = \sum_c (\text{row}_c)(\text{col}_c)$ · the chance-correction $\kappa = \frac{P_o - P_e}{1 - P_e}$ · why subtracting P_e removes lucky agreement · Landis–Koch bands and the $\kappa > 0.6$ publish floor · the kappa paradox under class imbalance.

Internal learning material. Numbers are reproducible from Appendix A. v1.0

1 The setup: two reviewers, fifty trajectories, one yes/no

Module 3.5’s first quality gate is *inter-annotator agreement* (IAA): if your two reviewers do not agree with each other, the labels they produce are noise, and any headline you build on them is noise too. The simplest case is two annotators, each giving every item one of two labels. Here: two reviewers, $N = 50$ agent trajectories, the single question “*is this resolution acceptable?*” — label **good** or **bad**.

Cross-tabulating their answers gives a 2×2 table. Each cell counts items: rows are what **Annotator A** said, columns are what **Annotator B** said. The two *diagonal* cells are where they agreed; the two *off-diagonal* cells are where they clashed.

	B: good	B: bad	A total
A: good	28	3	31
A: bad	4	15	19
B total	32	18	50

The quantities. These counts are illustrative but chosen round so every sum below follows on paper. The mechanics are identical at any size — only the cells change.

Quantity	Symbol	Value here
Items both called good (diagonal)	a	28
A good, B bad (off-diagonal)	b	3
A bad, B good (off-diagonal)	c	4
Items both called bad (diagonal)	d	15
Total items	$N = a+b+c+d$	50

Notation. A *marginal rate* is how often one annotator used a label, ignoring the other: A called things good 31/50 of the time, B called them good 32/50. P_o is observed agreement; P_e is the agreement we would expect if both annotators just threw their own labels down *at random* at those marginal rates. We carry four decimals throughout.

Intuition

Raw agreement always looks impressive, because some agreement is free. If two reviewers both say “good” to 90% of items out of sheer habit, they will agree most of the time *even with their eyes closed*. κ asks a sharper question: of the agreement that was *not* guaranteed by their label habits, how much did they actually achieve? It is raw agreement with the luck refunded.

2 Step 1 — observed agreement P_o

Observed agreement is just the fraction of items the two annotators landed on together — the diagonal of the table over the total:

$$P_o = \frac{a + d}{N} = \frac{\text{they agreed}}{\text{all items}}. \quad (1)$$

For our table, the diagonal is $a + d = 28 + 15 = 43$ items, so

$$P_o = \frac{28 + 15}{50} = \frac{43}{50} = 0.8600.$$

That is the “86% of items agreed” headline. It is true — and, on its own, almost useless, because it does not yet know how easy that 86% was to get by chance. ✓ ($43/50 = 0.86$.)

3 Step 2 — chance agreement P_e from the marginals

Now we compute how much these two would have agreed *by luck alone*. Suppose each annotator keeps their own label habit but assigns labels independently of the other — A says “good” with probability equal to A’s marginal rate, B independently with B’s marginal rate. The chance they *both* say good is the product of those two rates; likewise for both saying bad. Summing over the labels:

$$P_e = \sum_{c \in \{\text{good}, \text{bad}\}} \left(\frac{\text{row}_c}{N} \right) \left(\frac{\text{col}_c}{N} \right). \quad (2)$$

Read the marginals straight off the table. A called things good 31 times and bad 19; B called them good 32 times and bad 18:

$$\frac{\text{row}_{\text{good}}}{N} = \frac{31}{50} = 0.62, \quad \frac{\text{col}_{\text{good}}}{N} = \frac{32}{50} = 0.64, \quad \frac{\text{row}_{\text{bad}}}{N} = \frac{19}{50} = 0.38, \quad \frac{\text{col}_{\text{bad}}}{N} = \frac{18}{50} = 0.36.$$

So the two chance terms are

$$P_e = \underbrace{0.62 \times 0.64}_{\text{both “good” by luck}} + \underbrace{0.38 \times 0.36}_{\text{both “bad” by luck}} = 0.3968 + 0.1368 = 0.5336.$$

Mini-example — this operation in isolation

Why *multiply* the marginals? Independence. If A says “good” 62% of the time and B says “good” 64% of the time, and they are not looking at each other’s answers, then “both say good” happens $0.62 \times 0.64 = 0.3968$ of the time — the same rule as two independent coin flips. Add the “both bad” luck and you get $P_e = 0.5336$: with these label habits, two *random* reviewers would already agree on 53% of items, having read nothing. That free 53% is exactly what $P_o = 0.86$ must be judged against.

4 Step 3 — the chance-correction, and κ

Cohen’s idea: measure agreement not against 0, but against the chance floor P_e . The *room for non-chance agreement* is $1 - P_e$ (the gap between certain agreement and pure luck). κ is the fraction of that room the annotators actually covered:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (3)$$

The numerator $P_o - P_e$ is the agreement *above chance*; the denominator $1 - P_e$ is the agreement that was *available* above chance. Their ratio is normalised: $\kappa = 1$ means the annotators took all the available room (perfect agreement), $\kappa = 0$ means they did no better than luck, and $\kappa < 0$ means they did *worse* than two random reviewers (systematic disagreement). Plugging in:

$$\kappa = \frac{0.8600 - 0.5336}{1 - 0.5336} = \frac{0.3264}{0.4664} = 0.6998 \approx \mathbf{0.70}.$$

✓ Check: $0.3264/0.4664 = 0.6998$. The 86% raw agreement, once luck is refunded, is worth a κ of 0.70 — “substantial,” and just over the publish floor. The 16 percentage points between P_o and κ are precisely the lucky agreement we stripped out.

Intuition

Same numerator, different denominator, is the whole trick. $P_o - P_e = 0.326$ is “how far above luck they got.” Dividing by $1 - P_e = 0.466$ asks “out of how far they *could* have gotten.” A study where chance agreement is already 0.95 leaves almost no room above it, so even near-perfect raw agreement yields a modest κ . That is not a bug — it is the metric refusing to credit agreement that the label distribution was handing out for free. We meet its sharpest form in Section 7.

5 Step 4 — three categories, same machine

Nothing about Equations (1)–(3) is special to two labels. Suppose the reviewers grade each trajectory **good** / **borderline** / **bad**. The table is now 3×3 ; the diagonal still holds the agreements, the marginals are still row and column totals, and P_e still sums a product per category.

	B: good	B: border	B: bad	A total
A: good	20	3	1	24
A: border	4	12	2	18
A: bad	1	2	5	8
B total	25	17	8	50

Diagonal = $20 + 12 + 5 = 37$, so $P_o = 37/50 = 0.7400$. The three chance terms use the matching row/column rates:

$$P_e = \underbrace{\frac{24}{50} \frac{25}{50}}_{0.48 \cdot 0.50} + \underbrace{\frac{18}{50} \frac{17}{50}}_{0.36 \cdot 0.34} + \underbrace{\frac{8}{50} \frac{8}{50}}_{0.16 \cdot 0.16} = 0.2400 + 0.1224 + 0.0256 = 0.3880.$$

$$\kappa = \frac{0.7400 - 0.3880}{1 - 0.3880} = \frac{0.3520}{0.6120} = 0.5752 \approx \mathbf{0.58}.$$

✓ “Moderate” — below the publish floor. Splitting “good” into good/borderline gave the reviewers a harder, finer distinction; raw agreement fell to 0.74 and corrected agreement to 0.58. The same three lines of arithmetic scale to any number of categories.

6 Interpreting κ : the Landis–Koch bands

A raw κ needs a yardstick. The module uses the Landis–Koch (1977) bands — “not gospel, but the standard reference”:

κ range	Label	Action in a human eval
< 0	poor (worse than chance)	rubric inverted; stop
0.0–0.2	slight	guessing; re-train, re-write rubric
0.2–0.4	fair	still mostly noise
0.4–0.6	moderate	trend-spotting only; not a headline
0.6–0.8	substantial	publish — quote with κ alongside
0.8–1.0	almost perfect	gold-set quality

Why $\kappa > 0.6$ is the usual “trust this label set” floor. Below 0.4 the module says reject outright: the rubric is broken or the annotators are guessing. Between 0.4 and 0.6 the labels carry *some* signal but too much disagreement to put a number in front of an exec. At 0.6 the annotators are covering more than half the room above chance *and* the disagreements are small enough that aggregating across items averages them out. So 0.6 is where a label set stops being “one person’s opinion with noise” and becomes “a measurement two people would reproduce.” Our 2×2 study ($\kappa = 0.70$) clears it; the 3-category split ($\kappa = 0.58$) does not — quote it for trends, but tighten the rubric before it becomes a headline.

7 The kappa paradox — high agreement, low κ

The most dangerous failure of raw agreement is *class imbalance*. Suppose almost every trajectory is genuinely acceptable, so both reviewers say “good” nearly always. Consider $N = 100$:

	B: good	B: bad	A total
A: good	90	4	94
A: bad	4	2	6
B total	94	6	100

Raw agreement is excellent: $P_o = (90 + 2)/100 = 0.9200$. But both annotators say “good” 94% of the time, so chance agreement is enormous:

$$P_e = 0.94 \times 0.94 + 0.06 \times 0.06 = 0.8836 + 0.0036 = 0.8872.$$

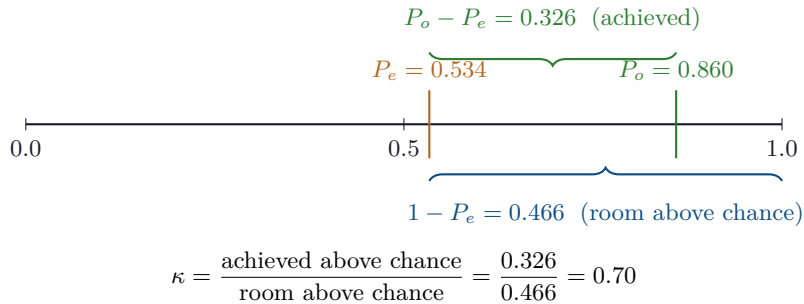
$$\kappa = \frac{0.9200 - 0.8872}{1 - 0.8872} = \frac{0.0328}{0.1128} = 0.2908 \approx \mathbf{0.29}.$$

✓ Ninety-two percent agreement, “fair” κ . The reviewers agreed on 92 items — but 88.7 of those were going to agree by label habit anyway, so they earned only 3.3 points of non-chance agreement out of 11.3 available.

Watch out

Never quote raw agreement on an imbalanced label. When one class dominates, P_e creeps toward 1, the denominator $1 - P_e$ collapses, and κ can be “fair” or even near zero while P_o reads 0.9+. The fix is structural, not cosmetic: balance the sample (over-sample the rare “bad” class for the IAA check), report κ *with* the marginal rates so the imbalance is visible, and — if your decision turns on the rare class — look at per-class agreement, not the pooled number. A 92% that hides a κ of 0.29 is exactly the kind of false confidence Module 3.5 exists to prevent.

8 The correction, drawn



Everything left of P_e is luck and does not count. κ is the green span (what the annotators earned) divided by the blue span (what was there to earn). The picture *is* Equation (3).

9 Kappa cheat-sheet

Observed agreement	$P_o = \frac{\text{diagonal}}{N} = \frac{a + d}{N}$ (2-class)
Chance agreement	$P_e = \sum \left(\frac{\text{row}_c}{N} \right) \left(\frac{\text{col}_c}{N} \right)$
Cohen's kappa	$\kappa = \frac{P_o - P_e}{1 - P_e}$
Range	$\kappa \leq 1$; 0 = chance, 1 = perfect, < 0 worse-than-chance
Publish floor	$\kappa \geq 0.6$ (substantial) to quote a headline
Paradox flag	one class dominant $\Rightarrow P_e \rightarrow 1$, κ collapses despite high P_o
Right metric when	2 raters, 2–5 nominal classes (else Fleiss' κ / Krippendorff's α)

10 An honest word about these numbers

The cell counts here (28/3/4/15, the 3×3 grid, the 90/4/4/2 paradox) are illustrative round numbers, picked so every P_o , P_e , and κ follows on paper — exactly as the gold template used round token sizes no one would ship. Your real tables come from your own reviewers and rubric, and your κ will move with them. What is *not* illustrative is the structure: κ is always observed-minus-chance over one-minus-chance, chance is always the product of marginals summed over labels, and the paradox always bites when a class dominates. Change the cells and the number moves; the chance-correction does not. That correction is the thing to carry into every human eval — it is why Module 3.5 gates on κ , not on raw agreement, before any number is allowed out the door.

Appendix A — Reference implementation

Every number in this document is produced by the program below (pure standard library — no dependencies). Run it to reproduce the 2×2 study, the 3-category study, and the paradox; change the tables to score your own annotators.

```

1  # cohens_kappa.py -- reproduces every number in "Cohen's Kappa, by Hand".
2
3  def cohen_kappa(table):
4      """table[i][j] = # items Annotator A labeled class i and B labeled class j."""
5      K = len(table)
6      N = sum(sum(row) for row in table)
7      agree = sum(table[i][i] for i in range(K)) # diagonal
8      Po = agree / N # observed agreement
9      row = [sum(table[i]) / N for i in range(K)] # A's marginal rates
10     col = [sum(table[i][j] for i in range(K)) / N # B's marginal rates
11            for j in range(K)]
12     Pe = sum(row[c] * col[c] for c in range(K)) # chance agreement
13     kappa = (Po - Pe) / (1 - Pe)
14     return N, agree, Po, Pe, kappa
15
16 def band(k):
17     for lo, hi, name in [(-1,0,"poor"),(0,.2,"slight"),(.2,.4,"fair"),
18                        (.4,.6,"moderate"),(.6,.8,"substantial"),(.8,1.01,"almost perfect")]:
19         if lo <= k < hi:
20             return name
21 
```

```
22 # --- 2x2 study: N=50, good/bad ---
23 T2 = [[28, 3], # A=good : (B=good, B=bad)
24       [ 4, 15]] # A=bad
25 N, ag, Po, Pe, k = cohen_kappa(T2)
26 print("2x2 N=%d diag=%d Po=%.4f Pe=%.4f kappa=%.4f (%s)" % (N, ag, Po, Pe, k, band(k)))
27
28 # --- 3-category study: good / borderline / bad ---
29 T3 = [[20, 3, 1],
30       [ 4, 12, 2],
31       [ 1, 2, 5]]
32 N, ag, Po, Pe, k = cohen_kappa(T3)
33 print("3cat N=%d diag=%d Po=%.4f Pe=%.4f kappa=%.4f (%s)" % (N, ag, Po, Pe, k, band(k)))
34
35 # --- the kappa paradox: high raw agreement, low kappa under imbalance ---
36 Tp = [[90, 4],
37       [ 4, 2]]
38 N, ag, Po, Pe, k = cohen_kappa(Tp)
39 print("para N=%d diag=%d Po=%.4f Pe=%.4f kappa=%.4f (%s)" % (N, ag, Po, Pe, k, band(k)))
40
41 # Expected output:
42 # 2x2 N=50 diag=43 Po=0.8600 Pe=0.5336 kappa=0.6998 (substantial)
43 # 3cat N=50 diag=37 Po=0.7400 Pe=0.3880 kappa=0.5752 (moderate)
44 # para N=100 diag=92 Po=0.9200 Pe=0.8872 kappa=0.2908 (fair)
```

— end of document —